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Manuscript Title: Integrating high-intensity physical activity breaks into senior school lessons: Teachers' perceptions of the 'Burn 2 Learn' program

INTRODUCTION

Less than 20% of adolescents satisfy physical activity guideline recommendations, with the lowest levels of activity observed among older adolescents (15-19 years) (Guthold et al., 2020). This stage of life is often characterized by significant changes that may influence physical activity participation, such as changes in employment/education status, increasing independence, and changing social roles (Beauchamp et al., 2018). This is also a crucial period for the development of mental ill-health, with approximately half of mental disorders emerging by the age of 18 (Solmi et al., 2022). Academic-related stress appears to be a major contributor, particularly during senior secondary school (i.e., the final two years of school (Wuthrich et al., 2020). There is compelling evidence that physical activity may help buffer the effects of stressful life events (Mücke et al., 2018), and contribute to positive mental health via a range of mechanisms (Lubans et al., 2016). As such, providing adolescents with regular opportunities to be physically active may help alleviate the elevated levels of stress and anxiety experienced by adolescents worldwide.

Schools are widely regarded as ideal settings for physical activity promotion. Physical education is the primary means of engaging adolescents in physical activity during the school day. However, in many countries, physical education is not a compulsory component of the senior school curriculum (i.e., Grades 11 and 12). (Beauchamp et al., 2018). Rather, curriculum time is redirected towards other academic subjects with a focus on maximizing learning time and academic performance. This is mainly due to the high stakes testing that occurs in the final years of secondary school (Wuthrich et al., 2020), which places considerable pressure on schools, teachers, and students. Classroom physical activity breaks

represent a promising strategy for integrating physical activity into the school day (Daly-Smith et al., 2018; Watson et al., 2017). However, most of this research has been conducted in primary schools, with limited evidence of effectiveness in secondary school settings (especially among senior school students) (Masini et al., 2022). There are unique barriers to delivering classroom activity breaks in secondary schools which may explain why there is limited research in this area (Fenesi et al., 2022). Schools may be reluctant to integrate physical activity into the school day, for fear of the academic costs of reallocating lesson time towards physical activity. Other barriers, such as lack of space and time, teacher attitudes, and the daily structure of high schools, present significant challenges for researchers designing interventions for these settings (Naylor et al., 2015).

High-intensity interval training (HIIT) may provide a time-efficient solution for classroom activity breaks in secondary schools. HIIT is a potent form of exercise that can produce a range of benefits for adolescents (Costigan et al., 2015; Leahy et al., 2020). While HIIT has become increasing popular in the school setting, it has rarely been delivered as a classroom activity break (Duncombe et al., 2022; Leahy et al., 2024). The most common application of HIIT in secondary schools is an adjunct to physical education classes (i.e., HIIT performed as part of a warm-up). Given the increasing time pressures associated with senior secondary schooling (and no compulsory physical education), implementing physical activity interventions with this population can be challenging. As such, we designed a high-intensity activity break intervention for senior school students, known as Burn 2 Learn (B2L).

With growing quantitative evidence to support the utility of HIIT in schools, qualitative evaluations can provide contextually specific information that may help improve our understanding of program implementation. Due to the complexities of schools, physical activity interventions are vulnerable to poor implementation, particularly as studies progress from efficacy to effectiveness trials (Lane et al., 2021). To help inform future efforts to

deliver at scale, there is a need to understand the perspectives of those directly involved in program implementation (i.e., classroom teachers) (Fenesi et al., 2022). Qualitative research can help identify facilitators and barriers to implementation, which are central to guiding intervention adaptations, implementation, and scale-up. Therefore, the purpose of the current study was to use a qualitative research methodology to examine teachers' perceptions of delivering the B2L physical activity intervention.

METHODS

Study design and participants

Participants for the current study were **senior** secondary school teachers who participated in the B2L cluster RCT (Leahy et al., 2019). Briefly, 20 secondary government schools from New South Wales, Australia were recruited and randomized to either the B2L intervention (n=10 schools), or a wait-list control condition (n=10 schools). Following the study period, teachers who were responsible for delivery of the B2L intervention in their school were invited via email to participate in a semi-structured interview to discuss their experiences of the program. Teachers were offered a $50 gift card if they agreed to participate. Of those invited, twelve teachers (63%) from eight schools agreed to participate in a 30-minute interview conducted by a member of the research team (MB), who was not involved in the delivery of the intervention. Ethics approval was provided by the human research ethics committee of the University of Newcastle, Australia (H-2016-0424) and the NSW Department of Education (SERAP: 2017116). Teachers and school principals provided written informed consent for the study.

Intervention description and setting

A detailed description of the intervention and RCT findings can be found elsewhere (Leahy et al., 2019; Lubans et al., 2021b; Mavilidi et al., 2021; Valkenborghs et al., 2022). Briefly, B2L

is a school-based physical activity intervention designed for older adolescents in senior schooling (i.e., Grades 11 and 12). Students participated in high-intensity physical activity breaks that could be performed with minimal to no equipment (i.e., jumping jacks, body-weight squats) (Figure 1). Activity breaks were designed to be completed in 10-20 minutes (including a warm-up/cool down). Teachers from intervention schools were provided with training, resources, and support strategies to facilitate the delivery of B2L sessions during academic lesson time. There were no restrictions on the subject area of teachers recruited as part of the study. The B2L intervention was delivered in three phases across two cohorts (12-months per cohort). In Phases 1 and 2 (May-September [2018 cohort 1 and 2019 cohort 2]), teachers were asked to facilitate the delivery of 2-3 weekly activity breaks during academic lessons. During Phase 3, (October-April [2018/2019 cohort 1 and 2019/2020 cohort 2]), teachers could continue to deliver weekly school-based sessions if they desired. Twelve teachers participated in the current study (Table 1). Subjects that B2L sessions were delivered in were as follows: Personal Development, Health, and Physical Education (PDHPE) (six teachers), Maths (two teachers) Sport, Lifestyle and Recreation (SLR) (two teachers), Sports coaching (one teacher) and Community and Family Studies (CAFS) (one teacher).

In the context of our study, senior schooling refers to the final two years of secondary school education (i.e., Grades 11 and 12). In senior schooling, students typically focus their studies on more specialized subject areas as they prepare for post-school pathways (e.g., higher education, vocational training). Further, students are not required to complete senior schooling, and instead may choose to enter the workforce and pursue other pathways. Although there is some variability across states (Stylianou & Walker, 2018), Australian schools are required to provide students in primary (Kindergarten-Grade 6) and early secondary schooling (Grades 7-10) with 150 mins of planned physical activity across the school week. This is usually addressed by compulsory physical education and school sport,

however, this is not a mandatory component of the senior school curriculum, as there is a greater focus on academic attainment and university/college entrance examinations.

Theoretical and implementation frameworks

Several frameworks were used to guide the B2L intervention. First, physical activity interventions targeting children and adolescents have been largely unsuccessful. The Theory of Expanded, Extended and Enhanced Opportunities (TEO) suggests that young people's physical activity is largely determined by adults (Beets et al., 2016). Specifically, the introduction of entirely new opportunities to be active (i.e., expansion) is one mechanism to improve youth physical activity. The B2L intervention adopts an application of the expansion mechanism, whereby a sedentary activity (e.g., seated classwork) is replaced with high-intensity activity breaks during lesson time. Second, B2L sessions were designed with reference to self-determination theory, to satisfy students basic psychological needs of autonomy, connectedness, and relatedness (Deci & Ryan, 1985). These basic psychological needs were operationalized through the SAAFE (Supportive, Active, Autonomous, Fair, Enjoyable) teaching principles (Lubans et al., 2017), and were introduced to teachers as part of the professional development they received. For example, to satisfy students' autonomy teachers would provide students with opportunities for choice within sessions (e.g., music playing). Student competence was satisfied by positive and specific feedback provided by teachers. Lastly teachers were encouraged to promote supportive behaviour among students (e.g., high fives) to support student relatedness.

Program implementation was guided by the Consolidated Framework for Implementation Research (CFIR), and we used a range of strategies used to maximise implementation (Damschroder et al., 2009). Specifically, we included implementation strategies across the five CFIR domains: (1) intervention characteristics (e.g., intervention

adaptability and acceptability, design quality and packaging of intervention resources), (2) outer setting (e.g., alignments with external policies and incentives), (3) inner setting (e.g., promoted to schools as a strategy to improve cognitive function and mental health), (4) characteristics of individuals (e.g., professional development for teachers to enhance self-efficacy), (5) implementation process (e.g., external change agent support and action planning). A detailed overview of the specific implementation strategies, outcomes, and determinants have been reported elsewhere. For the current study, program implementation was assessed by the number of self-reported sessions delivered by teachers. To ensure qualitative evaluation encompasses teacher perceptions across varied levels of implementation success, we organised implementation status as follows: High implementation was defined as delivering the prescribed number of sessions during the intervention period (≥ 28 sessions), moderate implementation was defined as delivering 24-27 sessions, and low implementation was defined as delivering ≤ 23 sessions. Most teachers (n = 8, 66%) delivered the program with high fidelity (i.e., delivered the prescribed number of sessions) while few teachers achieved moderate (n = 2, 17%), or low (n = 2, 17%) implementation.

Study measures

The conduct and reporting of this study adheres to the guidelines outlined in the Standards for Reporting Qualitative Research (O'Brien et al., 2014). We utilized a semi-structured interview methodology, as it allowed individual teachers to describe their experiences with the B2L intervention and identify contextual facilitators and barriers to implementation. A semi-structured interview framework was designed by the research team to help facilitate discussion and reflection on four broad topics that aligned with the CFIR domains: (1) overall program experiences, (2) program facilitators/barriers, (3) program improvements/alternative ideas, and (4) concluding comments (see Table 2). Semi-structured interviews were

conducted to explore teachers' experiences of facilitating the delivery of the B2L intervention in their school. Due to COVID-19 restrictions, interviews were administered online via Zoom, taking place between August and November 2020. One male research assistant, who was not involved in the intervention conducted all interviews (MB). Teachers were reminded at the beginning of the interview, that they may withdraw at any time without needing to provide a reason. At the completion of the interview, teachers were asked if they wanted to review any part of the recording to confirm or delete the contents that were contributed during the interview process.

Trustworthiness

We followed Shenton's provisions to enhance trustworthiness of data (Shenton, 2004). To support credibility of the findings, authors engaged in reflective discussion during every phase of the project, including interview question development (peer scrutiny), approach, data collection (reflective commentary, research methods) and analysis (peer debriefing). One male academic with previous teaching and qualitative data collection experience, who was not involved in the intervention delivery and had minimal prior contact with study participants, conducted all interviews (MB). Analyses were conducted by an external qualitative researcher (VH), with extensive experience and who was not involved in intervention development or delivery (background and experiences of the researcher). At the completion of the interview, participants were asked if they wanted to review any part of the recording to confirm or delete the contents that were contributed during the interview process (member checks). No teachers opted to review their interview transcription. Our study was informed by prior work investigating the 'scale-up penalty' in school-based research (Lane et al., 2021). To help address this issue, there is a need to appraise the scalability and acceptability of interventions, by understanding the perspectives of those directly involved in implementation. In addition to this conceptual foundation, data were gathered from a diverse

sample of teachers from different schools and socioeconomic backgrounds, supporting the transferability of our findings. To support dependability and confirmability, we provide a transparent outline of the process used within the study such that other researchers could repeat the work, such as triangulating participant interview data with self-reported implementation data.

Analyses

Audiotaped Zoom interviews were conducted and transcribed verbatim. Analyses were conducted using Nvivo 11 by an external qualitative researcher (VH), who was not involved in the intervention and had no previous contact with study participants. Template analysis was used for analysing the qualitative data. This analytic method is a variation of thematic analysis which is deemed particularly useful for research in which it is desirable to utilize a priori themes in forming initial coding schemes (King, 2012), in this case the CFIR domains. On this basis, an initial 'template' of domains was constructed to which teacher interview data were coded. This was expanded and refined further with the incorporation of in vivo codes until all relevant data were coded. A final version of the coding template was applied to the entire dataset. A priori themes were modified, redefined, or discarded to adequately align with the meanings and stories told in the data. Verbatim quotes from participants which vividly represented the themes were extracted and presented throughout. To protect participants' privacy, the names of teachers and schools involved were replaced with pseudonyms during transcription. School Index of Community Socio-Educational Advantage (ICSEA) percentile and quartile values were used to provide a measure of relative educational advantage. A higher value indicates greater level of educational advantage. The mean ICSEA value of participating schools was in the 40th percentile (range 5th-90th percentile), meaning that on average, schools included in the study were more educationally advantaged than 40% of schools nationally.

RESULTS

The template analysis of data identified four broad themes: (1) Appeal and motivation, (2) Barriers and facilitators of implementation, (3) Program impact, and (4) Sustainability of B2L in schools.

Theme 1: Program appeal and motivation to deliver B2L in school

A focus on student well-being was an important catalyst for joining the program

School and teachers focus on student well-being. One of the more implicit but prevailing motivations for joining was an existing school focus on student mental health and well-being which aligned with program objectives;

> *We just really saw the benefits and that it could have an impact on our students. I think we've got a big emphasis on well-being here at our school, and I thought that there might be some benefits not only the fitness, but also to that well-being aspect. (Participant 3, female, high implementation, SLR teacher)*

However, it was not only values and support of schools and executive which motivated teachers to join, but importantly teachers themselves alluding to having had existing positive attitudes towards physical activity and held beliefs about mental and physical health, which largely were confirmed by the program;

> *I've always agreed that physical exercise really contributes to your concentration in academic tasks. And I've always believed that anyway. So, I think it kind of aligned with what I already believe, so that's why I was keen to sort of participate (Participant 8, female, high implementation, mathematics teacher).*

Perceived benefits to students. Several teachers also spoke about having perceived one of the overarching benefits of the program being that it allowed students to witness first-hand the important link between physical, cognitive, and emotional health;

> *It was a good way to show them, like really demonstrate to them, what sort of effect physical activity can have on your mental health and physical health as well and how quickly those sorts of things can be affected (Participant 6, male, high implementation, sport coaching teacher).*

This was seen by many as particularly salient for senior students who not only were perceived as potentially 'compromised' in their physical health and fitness due to changes to mandates for involvement in sport in senior years, but also due to considering academic performance as separate from physical health, and indeed often mutually exclusive;

> *I think that it looked at improving students' mental health, working obviously in collaboration with physical health and cognitive health as well. A lot of our programs that we do are for junior students, in Grades 7-10 that we focus on and there's not too many proactive programs for senior students, so I think that's definitely an area that excited us (Participant 12, female, high implementation, CAFS teacher).*

Alternatively, few teachers specifically mentioned the potential of calming and focusing students in the classroom, as having been a catalyst for joining the B2L;

> *Definitely the ability to focus. It's the fact that it helps them calm down when they do enter the classroom (Participant 11, female, low implementation, mathematics teacher).*

Theme 2: Barriers and facilitators of implementation

Facilitators of delivery

Teacher involvement and attitudes. A clear facilitator of program implementation was teacher attitudes toward physical activity and its benefits to students beyond physical health and fitness;

> *I saw the value of the students participating in it, so it wasn't a burden to go out and do our B2L sessions. I felt that it was going to benefit them. It was just another part of what we were doing, and it became a regular feature of our class (Participant 1, male, high implementation, PDHPE teacher).*

Feeling equipped to deliver. Almost all teachers talked of the program training, resources and support by the research team as being highly beneficial for successful implementation and delivery. Teachers felt well prepared and well equipped to commence B2L once they had attended training;

> *[The research support] was fantastic. Though they were great, the training was really good as well. I felt like that equipped us well to implement the program itself. (Participant 12, female, high implementation, CAFS teacher).*

Addressing student barriers to participation. One of the common threads that emerged in the interviews were the strategies used by teachers to facilitate implementation and minimize barriers to participation. One such strategy included flexibility in school uniform policy allowing participation regardless of uniform;

> *Most of our kids had the option on days where we did B2L, to wear their sports uniform, or they could wear their school uniform. I think having a little bit of flexibility there and just the focus on them participating, no matter what mood they're in or what they're wearing, was a huge benefit (Participant 9, male, high implementation, PDHPE teacher).*

Barriers to delivery

The most common reasons for diminishing motivation were due to the competing prioritization of academic content, and students' perception of relevance in subjects without a focus on physical activity.

> *Our students were really committed to it and we made it work. But at the end of year 11 they were starting to be really concerned about how it would impact their studies in year 12. And we did try to continue, at a reduced rate in year 12, but found that the assessment requirements in other subjects really made it very difficult for us as the kids began the year 12 course (Participant 5, female, moderate implementation, PDHPE teacher).*

Contextual factors as impediments. Classroom proximity to an area suitable for group physical activity was also seen as a barrier by several teachers, and was seen to consume an inappropriate amount of time which competed with lesson content requirements;

> *Where my classroom's situated, to get down to an area where we could have completed the exercises was a bit of a pain. And getting all the equipment set up took a bit longer than I would have thought. (Participant 8, female, high implementation, mathematics teacher).*

Student perception of relevance. An interesting barrier to delivery was mentioned by a few teachers and most prominently by those teaching into subject areas not traditionally seen to involve physical activity (e.g., Mathematics). They voiced a certain degree of resistance from students to participate in B2L sessions due to the lack of perceived relevance to learning content in that subject area.

> *The behaviour is what concerns me of the students. They can't really focus inside a classroom, let alone if I take them out, do an activity and come back in the room. Being maths and then going out and doing it, that would just sort of like, "Why are we*

doing exercise in maths?" (Participant 8, female, high implementation, mathematics teacher)

Theme 3: Program impact

Student impact

Classroom focus and concentration. The vast majority of teachers expressed having witnessed the positive impact of the program on students' concentration and focus in the classroom.

I think just those active minutes are really important for our students. I think the increase in their focus was also quite evident after they'd completed the session. They are a lot more awake. (Participant 10, female, moderate implementation, PDHPE teacher).

Student insight into the link between mental health and exercise. Many teachers noted that students themselves became aware of their own improvements in concentration and general well-being;

The conversations that they were having and the talk that they were having, even going home and doing it, getting family members involved, and I think they could see the benefits and they could see the impacts that it was having on them as well. (Participant 3, female, high implementation, SLR teacher).

This was seen as one of the real strengths of the program, this facilitation of students' own insight into the positive effects that physical activity can have on mental health and cognition;

Personally, I know how important that is but to try and instil it in kids, we can stand at the front of the classroom and say, "Physical activity can have an effect on your mental health," but until they actually participate in it and see the effect of it, it's

311 *really hard to explain to them (Participant 6, male, high implementation, sport*
312 *coaching teacher).*

313 One teacher spoke of students' profound acknowledgement of this link, and use of the
314 principles learnt during B2L as they prepared for their university entrance exams;

315 *I would get them to come in and then reflect 10 minutes after the exercise on how they*
316 *were feeling. They very quickly began to realize the value of what physical activity,*
317 *(and) high intensity exercise does help them get into the zone where they could study.*
318 *And many of them commented later about understanding that link and making sure*
319 *that they are physically active in their preparation for their examinations.*
320 *(Participant 4, female, moderate implementation, PDHPE teacher).*

321 ***Impact on teaching practices***

322 Most teachers spoke of an increased reliance on using physical activity as a way to "recharge"
323 students' focus in class;

324 *If I have a sluggish class in theory lessons, I'll go all right jump us, let's go. And then*
325 *go outside and do a couple of minutes of some exercise and then come back and*
326 *refocus (Participant 5, female, moderate implementation, PDHPE teacher)*

327 ***Program reach and dissemination***

328 **Program dissemination.** One of the benefits that had been evident was that other staff not
329 directly involved in B2L implementation in their classes were aware of the program, its aims
330 and potential benefits; which appeared, in many instances, to have fostered more cooperation
331 and positive attitudes towards any changes to practices and routine which were required to
332 implement the program;

333 *After the staff presentation as well, I remember that there were teachers that came to*
334 *me and said that they really saw the benefits and they asked for a little bit of*
335 *information on it, and they asked for resources and things that they could use*
336 *themselves (Participant 3, female, high implementation, SLR teacher).*

337 **Theme 4: Sustainability of B2L in schools**

338 ***The future of B2L in schools***

339 **The B2L legacy.** Although many teachers did not continue delivering the program, several
340 teachers talked about having continued implementing certain elements or underlying ideas of
341 the program in a partial and or modified form;

342 *I've continued with my year 12 PDHPE class, but not in the same intensity as what*
343 *the program set out. We've used elements of it in year nine units for the new PDHPE*
344 *syllabus around mindfulness (Participant 6, male, high implementation, sport*
345 *coaching teacher).*

346 A few teachers had indeed been so pleased with the program and its benefits, that they were
347 planning to hand it over to senior students to deliver to the whole school community;

348 *Our aim was to actually have some of our students deliver lunchtime workshops to*
349 *interested students or staff members or before school sessions or after school.*
350 *(Participant 4, female, moderate implementation, PDHPE teacher).*

351 While others revealed plans to implement B2L in senior years to set students up with good
352 routines before their final year. One teacher even talked of the enthusiasm of students who
353 had participated in B2L in year 11 carrying over, as they had headed into the more
354 academically demanding year 12;

355 *We do my Year 12 who were originally in the program. I do think that I would run it*
356 *again depending on my class and my cohort, probably another Year 11 moving into*
357 *Year 12 for HSC (Participant 10, female, moderate implementation, PDHPE teacher).*

358 **DISCUSSION**

359 The aim of our study was to examine teachers' perceptions of delivering the B2L intervention
360 to older adolescents during academic lessons. To the best of our knowledge, this is the first
361 study to examine teachers' perception of delivering a high-intensity physical activity break
362 intervention among senior school students. Overall, teachers perceived the B2L program
363 favourably, and expressed the importance of physical activity for attaining mental health and
364 academic benefits. However, competing pressures of academic content, proximity to space to
365 support physical activity, and students' perception of relevance in traditionally academic
366 subjects (i.e., Mathematics) were barriers to implementation.

367 The primary driver for teachers' participation in the B2L program were the benefits of
368 physical activity for students' mental health, which is consistent with previous research
369 (Gammon et al., 2019). Mental health is likely to be a salient outcome for teachers and
370 students, as approximately one in six students experience 'very high' or 'extreme' levels of
371 psychological distress during their final years of schooling (Wuthrich et al., 2020). While
372 physically activity is widely accepted to optimize mental health, opportunities for engaging in
373 physical activity during the school day are limited during in senior schooling (Beauchamp et
374 al., 2018). To help improve 'buy-in' physical activity interventions targeting senior school
375 students should leverage teachers' willingness to improve students' well-being and motivate
376 students to engage academically (Cothran et al., 2010). Our findings echoed this statement as
377 teachers reported that the mental health focus of B2L was an important catalyst for them to
378 join the program. This is an important consideration for the design of future school-based

379 interventions, and there is the potential to align such programs with school policy initiatives
380 and/or frameworks (e.g., well-being framework) (Powell & Graham, 2017).

381 Many teachers also noted the students themselves had become aware of the positive
382 mental health effects of engaging in physical activity. Participation in acute physical activity
383 (particularly high-intensity) may stimulate neurobiological responses, in turn enhancing
384 feelings of euphoria (Lubans et al., 2016). Of note, the B2L program led to a reduction in
385 students' perceived stress and improvements in subjective vitality (Leahy et al., 2023;
386 Mavilidi et al., 2021), which may explain the positive effects that teachers described in their
387 interviews. Teachers also reported improvements in students' concentration and focus during
388 academic classroom activities following participation in B2L sessions, which is consistent
389 with prior research in junior secondary schools (Jørgensen et al., 2020; Lerum et al., 2021)
390 Interestingly, there appears to be a dose-response effect, whereby greater effects on
391 classroom behaviour are observed for interventions adopting higher intensity activity (Daly-
392 Smith et al., 2018). While most activity break interventions have prescribed moderate-to-
393 vigorous physical activity, some primary school interventions have adopted higher-intensity
394 activity, with significant improvements in on-task behaviour (Broad et al., 2023; Ma et al.,
395 2015). High-intensity activity may improve on-task behaviour to a greater extent than lower
396 intensity activity, as students are required to utilise more energy, therefore increasing their
397 level of fatigue and reducing the likelihood to be disruptive (Mavilidi et al., 2021). Another
398 possible explanation is that high-intensity activity moderates the effect on students' mood
399 (i.e., enhanced feelings of vigour and energy), thus making students more prepared to learn
400 (Lubans et al., 2021a). Together, these findings suggest that short (but intense) activity breaks
401 can be a potent stimulus to enhance classroom engagement.

402 Despite the perceived benefits, several teachers expressed the difficulty of
403 reallocating time away from academic content. **Senior** schooling typically culminates with

high stakes examinations that have implications for future education and employment opportunities (i.e., university admissions rankings) (Wuthrich et al., 2020). While these exams are a major source of stress in themselves, it is plausible that the increased academic workload in preparing students for these exams is also contributing to elevated distress (Wuthrich et al., 2020). Our findings are similar to those previously reported in junior secondary settings (Gammon et al., 2019; Jørgensen et al., 2020), highlighting the significant academic-related barriers that are likely to contribute to poorer implementation of classroom activity breaks in secondary schools. Acknowledging this, researchers have started to reconceptualise activity breaks as part of an overall approach to proactive classroom management and effective teaching. Lander and colleagues (2024) suggest activity breaks can be used to structure, transition, manage, energize, and learn. As such, if time is unable to be allocated from academic instruction, 'structured' or 'learning' activity breaks may be more suitable in these learning contexts. In other cases, teachers may consider the potential 'trade-off' between the time taken to administer activity breaks, and the benefits that activity breaks may lead to (e.g., enhanced focus and alertness). In the context of our study, teachers noted how B2L sessions 'energised' students, and therefore could be an effective strategy on days where students need to become refocused and reengaged in their learning.

Teachers also mentioned they received resistance from some students in performing physical activity in subjects that are not traditionally perceived to be active. Despite this, qualitative research highlights that both students and teachers value classroom physical activity breaks in subjects that do not traditionally rely on movement (Gammon et al., 2019; Masini et al., 2024; Warehime et al., 2019), however prominent barriers should be acknowledged. For example, a lack of classroom space and resistance from students have been previously cited as barriers to implementation (Gammon et al., 2019; Warehime et al., 2019), which are consistent with findings from our current study. To help overcome this

429 barrier activity breaks that are shorter in duration (e.g., <10 minutes), and have less
430 complexity (e.g., can be completed with large group in standard classroom, with no
431 equipment) may be required to improve implementation in secondary school settings.

432 The structure of the school day is a common barrier for secondary teachers when
433 implementing classroom physical activity breaks (Fenesi et al., 2022), which was reiterated
434 by teachers in the current study. Teachers only have a limited time each day with their
435 students (some as few as 40 minutes per lesson), which may lead to greater resistance when
436 asking them to redirect time away from their core academic activities. Although this is less of
437 a concern in primary schools, previous research has found that secondary teachers face
438 greater struggles with finding time to implement such interventions due to the structuring of
439 academic lessons (McMullen et al., 2014). As such, providing teachers with the opportunity
440 to pilot test strategies and action plan to overcome barriers is likely to be beneficial in
441 creating greater agency among secondary school teachers (Mulhearn et al., 2023). While lack
442 of time is a prevalent barrier to classroom-based activity, one area that may have less time
443 constraints is special and inclusive education (Leahy et al., 2021; Mazzoli et al., 2021).
444 Special education classes typically allow greater flexibility in the delivery of academic
445 content, and teachers spend most of the day with the same students (similar to primary
446 school), therefore mitigating a major barrier to activity breaks in secondary schools (Cothran
447 et al., 2010).

448 Professional development opportunities (both pre-service and in-service) are key to
449 enhancing teachers' self-efficacy regarding the use of classroom activity breaks, particularly
450 in subject areas that do not have a focus on physical activity (Benes et al., 2016; Mulhearn et
451 al., 2023). Teachers acknowledged the importance of the B2L professional development,
452 however, the majority of teachers had a physical education background, and therefore may
453 have been more confident to deliver classroom activity breaks. There is emerging evidence

454 that initial teacher education programs (including undergraduate and postgraduate programs)
455 can enhance pre-service teachers' confidence to deliver physical activity pedagogy,
456 particularly classroom activity breaks (Lander et al., 2023). While promising, these studies
457 were conducted with pre-service primary teachers, which is not surprising considering the
458 lack of physical activity breaks that occur in secondary schools (in comparison to primary
459 schools). As such, future research should examine whether embedding physically active
460 pedagogy in initial secondary teacher education programs can yield similar improvements in
461 teacher competence and confidence.

462 Our study offers a unique contribution to the field with several strengths. First,
463 research examining classroom-based physical activity in secondary schools is scarce,
464 particularly in senior secondary school. Second, classroom activity breaks were facilitated by
465 teachers (as opposed to researchers), which strengthens the ecological validity of this
466 approach. Third, the qualitative research methodology was undertaken to provide a more in-
467 depth understanding of teachers' perspectives of delivering classroom physical activity
468 breaks during academic lessons, complementing previous quantitative findings. Importantly,
469 this included teachers with low, moderate, and high implementation fidelity. Finally, an
470 external researcher conducted the qualitative analysis. There are also some limitations that
471 should be acknowledged**.** First, we did not collect information regarding years of teaching
472 experience and ethnic background of teachers which would provide greater context to study
473 findings. Second, qualitative data may have been subject to interview bias. While we
474 attempted to limit this bias by having all interviews conducted by a member of the research
475 team who was not involved in the delivery of the intervention, social desirability bias cannot
476 be discounted and may pose as a potential threat to credibility. Third, the sample size is also
477 relatively small, and most teachers had a background in physical education which limits the
478 generalizability of results. However, it is important to note delivery of the senior school level

Health and Physical Education curriculum does not involve practical participation in physical activity. Rather, students routinely engage in classroom-based theoretical content learning related to exercise science, biomechanics, and health promotion. Finally, although teachers discussed student benefits of the program, we did not conduct any interviews/focus groups with students to confirm these findings due to COVID restrictions.

CONCLUSION

Our findings suggest that teachers were supportive of classroom physical activity breaks, with perceived benefits to students' well-being and classroom engagement being the most salient reasons for implementation. Despite the short duration of activity breaks (i.e., 10 mins), competing time demands was the most prominent barrier to implementation during academic lessons. As such, researchers may need to explore alternative times during the school day (i.e., in the morning before classes), to provide students with opportunities to be active if curriculum time is unable to be redirected away from academic subjects.

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Table 1. Characteristics of interview participants and implementation status.

Participant	Sex	ICSEA percentile, (quartile)	Subject discipline	Implementation status
1	M	69 (Q2)	Personal Development, Health, and Physical Education	High
2	M	36 (Q3)	Sport, Lifestyle and Recreation	Low
3	F	36 (Q3)	Sport, Lifestyle and Recreation	High
4	F	79 (Q1)	Personal Development, Health, and Physical Education	Moderate
5	F	79 (Q1)	Personal Development, Health, and Physical Education	High
6	M	5 (Q4)	Sports Coaching	High
7	M	21 (Q4)	Personal Development, Health, and Physical Education	High
8	F	21 (Q4)	Mathematics	High
9	M	90 (Q1)	Personal Development, Health, and Physical Education	High
10	F	10 (Q4)	Personal Development, Health, and Physical Education	Moderate
11	F	10 (Q4)	Mathematics	Low
12	F	27 (Q3)	Community and Family Studies	High

Note: ICSEA quartile defined as Q1 (100-75), Q2 (74-50), Q3 (49-25), Q4 (24-1). High implementation ≥28 sessions, moderate implementation 24-27 sessions, low implementation ≤23 sessions. NR = not reported.

Table 2. Semi-structured interview questions.

Interview topic and questions	Alignment with CFIR domain/s
Program experiences	
1. Overall, what did you think about the B2L program?	Intervention characteristics, inner setting, characteristics of individuals, implementation process
2. How did your students respond to the B2L program?	Intervention characteristics, characteristics of individuals
3. What were your experiences with the face-to-face teaching provided in B2L?	Intervention characteristics, implementation process
4. Have you continued or are thinking of continuing the B2L program with a new cohort of students? If so, which year group of students will you run the program with?	Intervention characteristics, implementation process
5. What were your experiences with the B2L resources?	Intervention characteristics, inner setting
Facilitators/barriers	
6. Can you tell me why you decided to run the B2L program at your school?	Intervention characteristics, outer/inner setting
7. What potential benefits from the program do you think are important for your students?	Intervention characteristics, inner setting, characteristics of individuals
8. Did you notice an impact on students?	Intervention characteristics
9. What were the factors that helped you deliver B2L in your school?	Intervention characteristics, outer/inner setting, characteristics of individuals, implementation process
10. Did you have trouble motivating students to participate? If so, what strategies did you use to motivate students?	Inner setting, characteristics of individuals, implementation process
11. Did your motivation change over time? What strategies did you use to increase your motivation?	Inner setting, characteristics of individuals, implementation process
12. Were there any major barriers to implementing the B2L program in your school?	Intervention characteristics, outer/inner setting, characteristics of individuals, implementation process
13. What was your experience with the B2L WhatsApp group?	Implementation process

14. Did completing an action plan in the training workshop help with the implementation of the B2L program in your school?	Implementation process
Improvements/alternative ideas	
15. How can the B2L program be improved?	Intervention characteristics
16. Has participation in B2L changed the way you teach?	Individual characteristics
17. Can you think of a better strategy for increasing physical activity in senior school students?	Outer/inner setting
18. How can students be supported to engage in B2L sessions outside of class?	Outer/inner setting
19. Have you shared your knowledge/resources with other teachers who did not attend the workshop?	Inner setting, characteristics of individuals, implementation process
Concluding comments	
20. Have any attitudes of fellow staff members changed since you have implemented the B2L program? Has your school network increased promotion of the program?	Inner setting, characteristics of individuals
21. If you were to leave the school, who would take over in delivering the program? Have you passed on any of the resources (e.g., USB, Teacher Handbook) for others to utilize?	Inner setting, characteristics of individuals, implementation process
22. Has the B2L program become a part of your school's PDHPE/PASS/SLR curriculum?	Intervention characteristics, implementation process
23. Do you have any further ideas comments on anything that was not covered in the questions?	Intervention characteristics, outer setting, inner setting, characteristics of individuals, implementation process

Note: B2L = Burn 2 Learn; CFIR = Consolidated Framework for Implementation Research.

Figure 1. High-intensity activity break task cards.